B. P. Mandal College of Engineering, Madhepura

Bachelor of Technology

A Project Report On

Design and Development of 3D Product Advertisement of Noise Wireless Earbuds Using Blender

Submitted by:

Team Members:			
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Branch: 3D Animation and Graphic

Semester (Year): 7th sem (4th year)

Session: 2022–2026

Under the Guidance of

Prof. Md. Hafeezullah

BPMCE 2022–2026

Declaration

We hereby declare that the project report entitled
"Design and Development of 3D Product Advertisement of Noise Wireless Earbuds Using Blender"
submitted in partial fulfillment of the requirements for the award of the degree of **B.Tech** in the subject **3D Animation and Rendering** is a record of original work carried out by us during the academic session 2025–26.
We further declare that:

- This project work has not been submitted previously to any other university or institution for the award of any degree or diploma.
- The work presented in this report is based on our practical implementation and learning using Blender software.
- All references and external sources used during the development of this project have been properly acknowledged.
- The project has been completed under the guidance and supervision of our faculty mentor.

We take full responsibility for the authenticity and originality of the work presented in this report.

Date: _____
Place: _____

Acknowledgement

We would like to express our sincere gratitude to our respected project guide and faculty members for their continuous guidance, encouragement, and valuable suggestions throughout the development of this project titled

"Design and Development of 3D Product Advertisement of Noise Wireless Earbuds Using Blender."

Their expert supervision helped us understand the complete 3D production pipeline, including modeling, shading, lighting, rendering, and post-processing techniques. Without their constant support and motivation, the successful completion of this project would not have been possible.

We are also thankful to our department for providing the necessary resources and learning environment required to carry out this project effectively.

We would like to acknowledge the open-source Blender community and online educational resources that helped us improve our practical understanding of 3D product animation techniques.

Finally, we express our heartfelt gratitude to our friends and classmates for their support, cooperation, and constructive feedback during the development of this project.

This project has greatly enhanced our technical knowledge, teamwork skills, and practical exposure to industry-level 3D animation workflows.

KETAN ANAND 22118128007
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Certificate

This is to certify that the project report entitled

"3D Product Advertisement of Noise Wireless Earbuds"

submitted by

Ketan Anand (Reg. No: 22118128007)
Pooja Kumari (Reg. No: 22118128009)
Ladli Kumari (Reg. No: 22118128010)

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech)** in **3D Animation and Graphics** for the academic session **2025–2026**, is a record of the work carried out by them under my supervision and guidance.

To the best of my knowledge, this project work has not been submitted elsewhere for the award of any other degree or diploma.

This project demonstrates the practical implementation of **3D modeling, shading, lighting, animation, rendering, and post-processing using Blender** for creating a product advertisement.

Mentor's Signature:

Prof. Md. Hafeezullah
Mentor

Head of Department's Signature:

Prof. Murlidhar Prasad Singh
HoD, CSE

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CHAPTER 1:
Introduction

In the modern digital era, product advertising has significantly evolved from traditional photography to computer-generated 3D visualization. Brands increasingly use 3D animation to present their products in a visually attractive and realistic manner. This project focuses on the creation of a professional-quality 3D advertisement of Noise Wireless Earbuds using Blender software.

The project was developed by a team of three members, where each member was assigned specific roles according to the 3D production pipeline. The entire workflow includes modeling, shading, texturing, lighting, animation, rendering, and post-processing.

The objective of this project was not only to create a visually appealing advertisement but also to understand and implement the complete 3D production pipeline used in professional studios.

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CHAPTER 2:

PROJECT TEAM STRUCTURE AND WORK DISTRIBUTION

Team Members

- Member 1 –
  Modeling & Asset Creation Lead
- Member 2 –
  Shading, Texturing & Lighting Artist
- Member 3 –
  Animation, Rendering & Post-Processing Artist

WORK DIVISION OVERVIEW

The project pipeline was divided into three main departments:

Modeling Department

Responsible for creating accurate 3D geometry.

Look Development Department

Responsible for materials, textures, and lighting.

Rendering & Post-Production Department

Responsible for animation, final rendering, and compositing.

This division ensured smooth workflow and efficient collaboration.

Objectives

The primary objective of this project titled

"Design and Development of a 3D Product Advertisement of Noise Wireless Earbuds Using Blender"

is to understand and implement the complete 3D production pipeline used in professional product visualization and advertising industries. The project aims to combine creativity with technical knowledge to produce a high-quality commercial-style advertisement using Blender software.

The specific objectives of this project are as follows:

1. To Understand the Complete 3D Production Pipeline

- The project aims to gain practical knowledge of the full 3D workflow, including:
- 3D Modeling
- Shading and Texturing
- Lighting Setup
- Camera Animation
- Rendering
- Compositing and Post-Processing
- Understanding this pipeline helps simulate real-world industry production methods.

2. To Develop Accurate 3D Modeling Skills

- Create a detailed and proportionally accurate 3D model of wireless earbuds and charging case.
- Apply proper topology techniques for smooth surfaces.
- Use modifiers efficiently for professional-quality geometry.

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3. To Apply Physically Based Rendering (PBR) Principles

- Learn how real-world materials interact with light.
- Create realistic plastic and metallic materials using the Principled BSDF shader.
- Control roughness, specular highlights, and reflections for product realism.

4. To Design Professional Studio Lighting

- Implement a three-point lighting system (Key, Fill, Rim light).
- Use HDRI environment lighting for realistic reflections.
- Understand the importance of light intensity, direction, and softness in product visualization.

5. To Create Cinematic Camera Animation

- Plan and execute smooth camera movements such as orbit, dolly-in, and reveal shots.
- Use keyframes and graph editor to refine animation curves.
- Apply depth of field to achieve cinematic focus effects.

6. To Perform High-Quality Rendering

- Use Cycles render engine for physically accurate results.
- Optimize render settings for balance between quality and performance.
- Reduce noise using denoising techniques.

7. To Apply Post-Processing Techniques

- Use Blender compositor for color correction and contrast adjustment.
- Enhance visuals using glare and exposure control.
- Export final advertisement in high-definition video format.

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8. To Enhance Team Collaboration Skills

- Divide work efficiently among three team members.
- Maintain organized workflow and communication.
- Simulate real-world production teamwork environment.

9. To Gain Industry-Relevant Experience

- Understand how 3D advertisements are created professionally.
- Build practical experience useful for careers in 3D animation, product visualization, and digital marketing industries.

10. To Improve Technical and Creative Skills

- Strengthen problem-solving abilities.
- Improve creativity in product presentation.
- Gain confidence in handling complete 3D projects independently.

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Scope of the Project

The scope of this project titled

"Design and Development of a 3D Product Advertisement of Noise Wireless Earbuds Using Blender"

covers the complete digital production workflow required to create a professional-quality 3D product advertisement. The project focuses on implementing industry-standard 3D animation techniques using Blender software and demonstrates practical application of modeling, shading, lighting, animation, rendering, and post-processing.

1. Technical Scope

The technical scope of this project includes the following components:

1.1 3D Modeling

- Creation of detailed and accurate 3D models of wireless earbuds and charging case.
- Application of proper topology and mesh optimization techniques.
- Use of modifiers such as Subdivision Surface, Bevel, and Mirror.

This ensures that the product geometry is smooth and suitable for high-quality rendering.

1.2 Shading and Texturing

- Implementation of Physically Based Rendering (PBR) principles.
- Creation of realistic plastic and metallic materials.
- Adjustment of surface properties such as roughness, specular, and metallic values.

This stage enhances realism and improves visual appeal.

1.3 Lighting Setup

- Designing a professional studio lighting environment.
- Application of three-point lighting system.
- Use of HDRI for realistic reflections and ambient illumination.

Lighting plays a major role in highlighting product features and surface quality.

1.4 Camera Animation

- Creation of cinematic camera movements such as orbit and dolly shots.
- Use of keyframes and graph editor for smooth motion.
- Application of depth of field for professional visual presentation.

This ensures that the advertisement appears dynamic and engaging.

1.5 Rendering

- Use of Cycles render engine for photorealistic output.
- Optimization of render samples and denoising.
- High-definition video output (Full HD).

Rendering converts the 3D scene into final visual output.

1.6 Compositing and Post-Processing

- Color correction and exposure adjustment.
- Addition of glare effects and contrast enhancement.
- Final export in video format suitable for advertisement.

Post-processing enhances the commercial look of the advertisement.

2. Educational Scope

This project provides practical learning in:

- Real-world 3D production pipeline
- Industry-standard product visualization workflow
- Team-based project management
- Problem-solving in rendering and animation

It bridges the gap between academic theory and industry practice.

3. Industry Scope

The techniques used in this project are widely applied in:

- E-commerce product advertisements
- Social media marketing campaigns
- Digital product launches
- Television and online commercials
- Brand promotion videos

3D product visualization is increasingly replacing traditional photography due to flexibility, cost efficiency, and creative control.

4. Future Scope

The project can be further enhanced by:

- Adding motion graphics and typography

- Incorporating sound design and background music
- Creating multiple product color variations
- Developing real-time versions using game engines
- Expanding into Augmented Reality (AR) product visualization

5. Limitations of the Project

While the project demonstrates a complete 3D workflow, it is limited to:

- A single product model
- Pre-rendered animation (not real-time)
- Basic compositing within Blender

Advanced simulation effects and large-scale production techniques were beyond the current scope.

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CHAPTER 3:

MEMBER 1 – MODELING & ASSET CREATION

Role and Responsibilities

Member 1 was responsible for:

- Reference collection
- Base mesh modeling
- High-detail modeling
- Topology refinement
- Modifier management
- Scene asset organization

Reference Study and Planning

Before modeling, detailed product references were collected. These references helped in understanding:

- Product proportions
- Surface curvature
- Edge smoothness
- Button placement
- Case opening mechanism

Base Modeling Process

Modeling began using primitive objects:

- Cylinder for earbud stem
- Sphere for curved cap
- Cube for charging case

Edit Mode was used to manipulate vertices and edges.

Subdivision Surface modifier was applied to smooth the geometry.

Bevel modifier was used to soften sharp edges.

Mirror modifier ensured symmetrical modeling.

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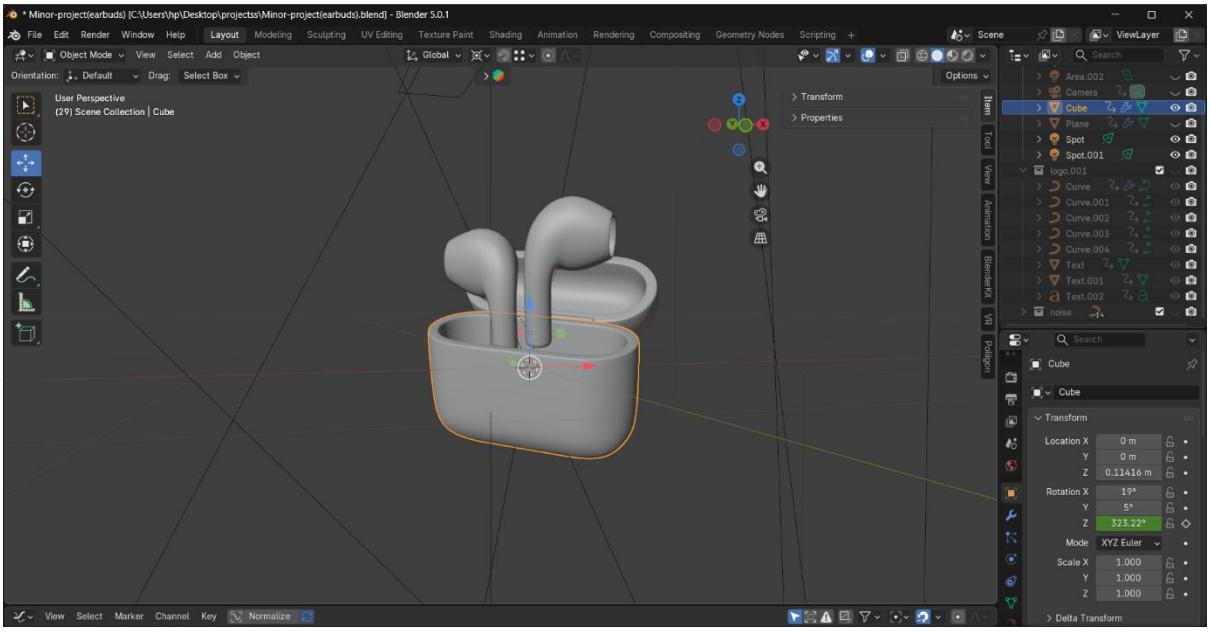


Fig.1

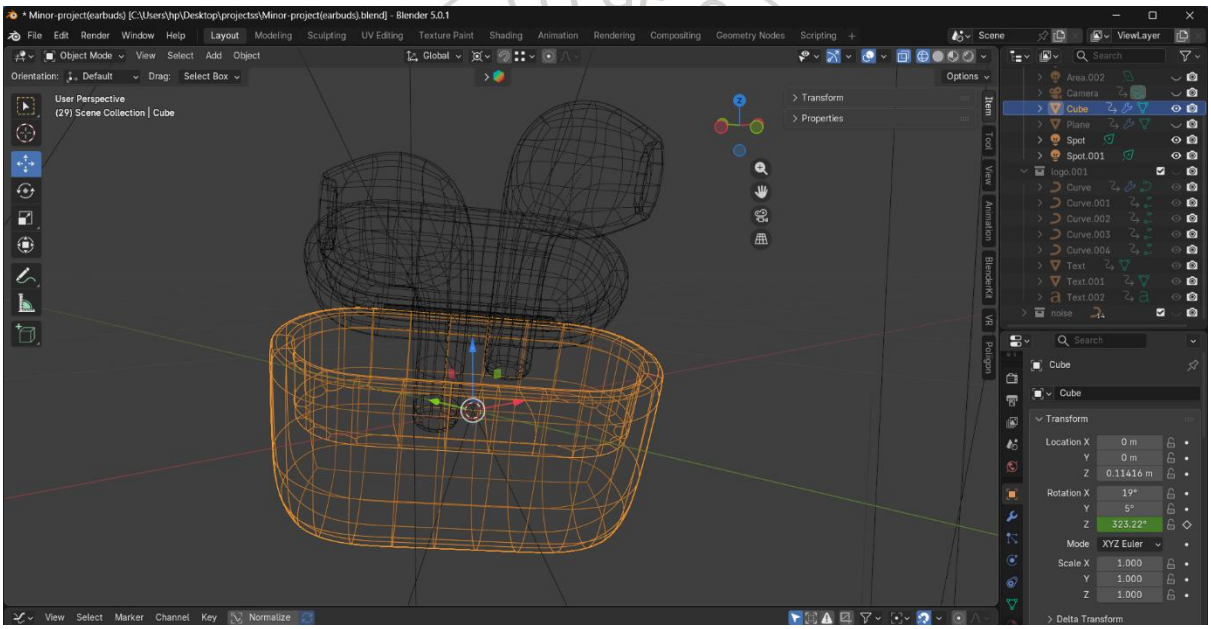


Fig. 2

Topology Management

Clean topology was maintained to ensure:

- Smooth shading
- Accurate reflections
- No rendering artifacts

Edge loops were strategically placed to control curvature.

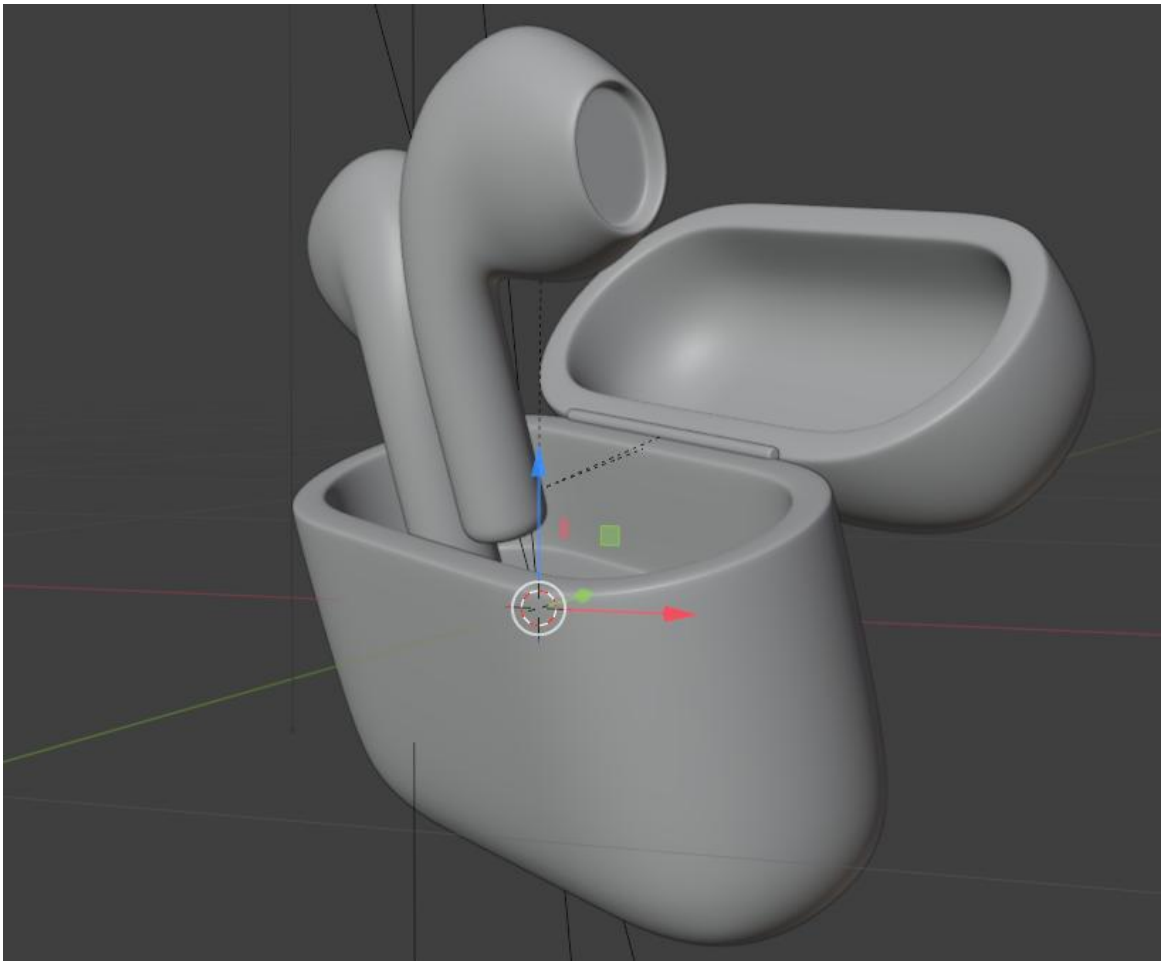
Proper topology is essential in product visualization because reflections reveal surface imperfections.

Final Model Output

The final 3D model consisted of:

- Two earbuds
- Charging case
- Lid mechanism
- Internal detailing

All parts were properly named and organized in the outliner panel.



CHAPTER 4:
MEMBER 2 – SHADING, TEXTURING & LIGHTING

Role and Responsibilities

Member 2 handled:

- Material creation
- Shader node setup
- Texture adjustments
- Reflection control
- Studio lighting setup
- HDRI environment setup

Physically Based Rendering (PBR)

The project used Physically Based Rendering principles.

PBR ensures that materials react realistically to light.

Key properties used:

- Base Color
- Roughness
- Metallic
- Specular
- Normal mapping

Material Creation

The Principled BSDF shader was used.

Plastic material setup:

- Low roughness
- Slight specular highlight

Metallic ring setup:

- Metallic value near 1
- Controlled roughness for reflection sharpness

Case interior:

- Matte plastic finish

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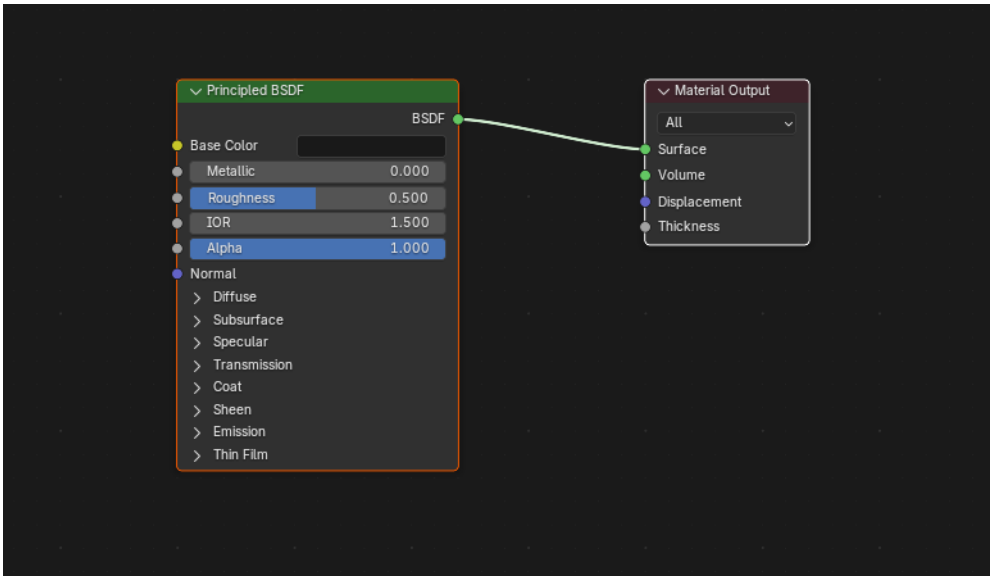

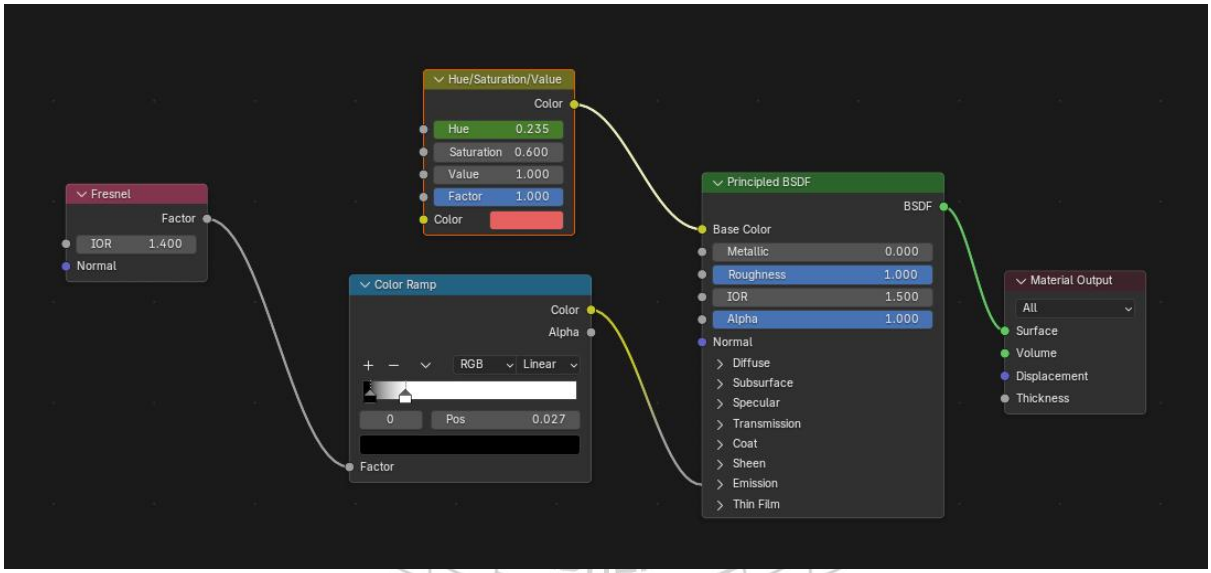

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Reflection and Surface Realism

Reflection is critical in product ads.
Small adjustments in roughness dramatically changed the realism.
Careful balancing ensured:

- No over-shiny plastic
- No dull surface

Lighting Design

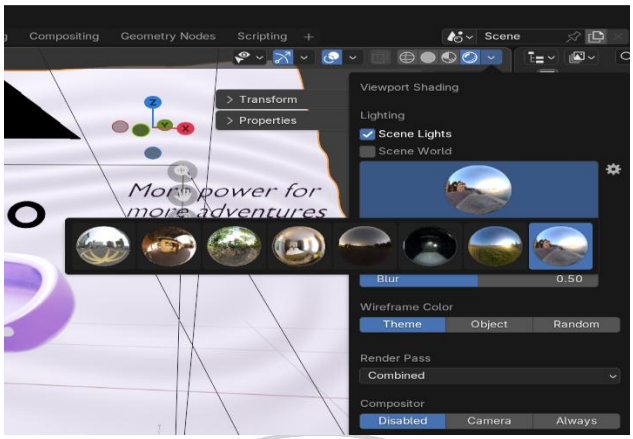
Lighting setup followed professional studio techniques.
Three-point lighting system:

1. Key Light – Main light source
2. Fill Light – Softens shadows
3. Rim Light – Highlights edges

HDRI was used for natural ambient reflections.

Lighting helped in defining:

- Product shape
- Surface texture
- Depth and contrast



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CHAPTER 5:

MEMBER 3 – ANIMATION, RENDERING & POST-PROCESSING

Role and Responsibilities

Member 3 was responsible for:

- Camera setup
- Keyframe animation
- Graph editor smoothing
- Render configuration
- Compositing
- Final export

Camera Animation

Professional product advertisements use cinematic camera movements.
Techniques used:

- Orbit rotation around product
- Slow dolly-in movement
- Close-up reveal
- Depth of Field for focus

Keyframes were added in timeline.

Graph Editor was used to smooth animation curves.

Rendering Process

Render Engine: Cycles

Cycles uses ray tracing to simulate realistic light behavior.

Render settings:

- Resolution: 1920 × 1080
- Frame rate: 24 fps
- Samples: 1000+
- Denoising enabled

Higher samples reduced noise but increased render time.

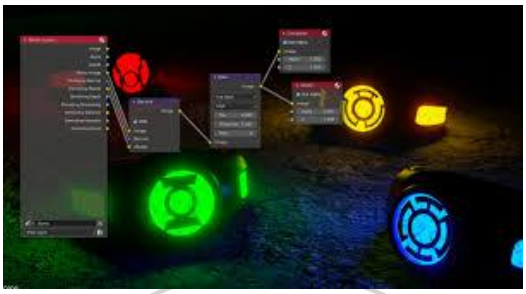
Noise Reduction

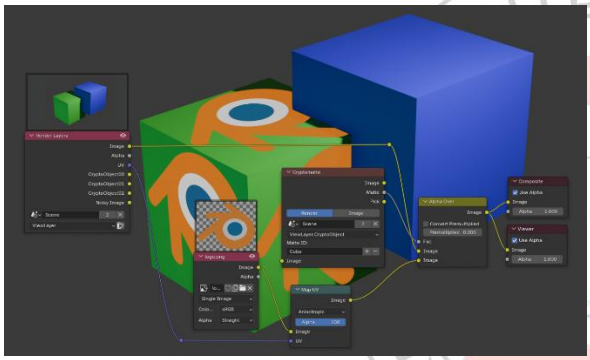
Noise is common in ray-traced rendering.

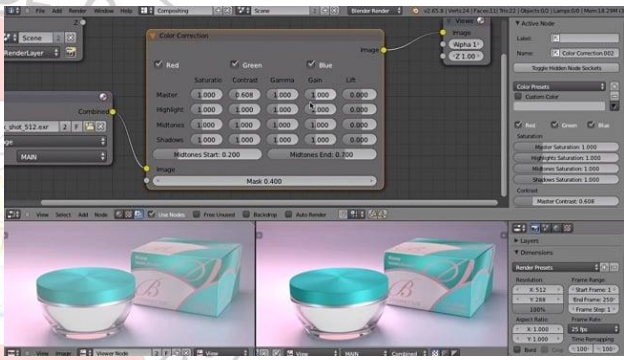
Solutions used:

- Increased samples
- Enabled denoiser
- Optimized light strength

Compositing and Post-Processing







Compositor nodes were used for:

- Color correction
- Contrast adjustment
- Glare effect
- Exposure balancing

Post-processing enhanced commercial appearance.

Final output exported as MP4 video.

CHAPTER 6:

TEAM COLLABORATION AND WORKFLOW

The team followed a structured workflow:

Step1 – Modeling completed

Step2 – Look development applied

Step3 – Lighting finalized

Step4 – Animation created

Step5 – Rendering executed

Step 6 – Post-processing completed

Regular discussions ensured consistency in visual style.

Assets were shared using common project files.

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CHAPTER 7:

CHALLENGES FACED BY TEAM

- Maintaining realistic plastic surface
- Reducing render noise
- Long rendering time
- Achieving smooth animation

These were solved through:

- Research and experimentation
- Render optimization
- Graph editor refinement
- Proper lighting adjustments

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CHAPTER 8:

LEARNING OUTCOMES

The team gained knowledge in:

- Professional 3D product workflow
- PBR material system
- Studio lighting principles
- Ray tracing rendering
- Post-production techniques
- Team collaboration in digital production

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CHAPTER 9:
CONCLUSION

The project successfully demonstrated the creation of a professional-quality 3D product advertisement. Each team member contributed significantly according to their assigned role. The collaboration ensured smooth workflow and high-quality final output.

The project strengthened our understanding of industry-standard 3D production techniques and prepared us for real-world digital content creation.

CHAPTER 10:

PRACTICAL IMPLEMENTATION AND COMPLETE WORKFLOW OF 3D EARBUD ADVERTISEMENT

1. REFERENCE COLLECTION & PROJECT SETUP

Step 1: Collect Reference Images

Before starting modeling, high-resolution reference images of Noise earbuds were collected from multiple angles:

- Front View
- Side View
- Top View
- Charging Case Open View

Step 2: Blender Project Setup

- Open Blender
- Delete default cube
- Set Units to Metric
- Enable Add-ons (if required)
- Save project file

2. MODELING THE EARBUD

Step 3: Create Earbud Body (Base Mesh)

- Add → Mesh → Cylinder
- Reduce vertices (16–24) for control
- Scale and rotate to match reference
- Enter Edit Mode

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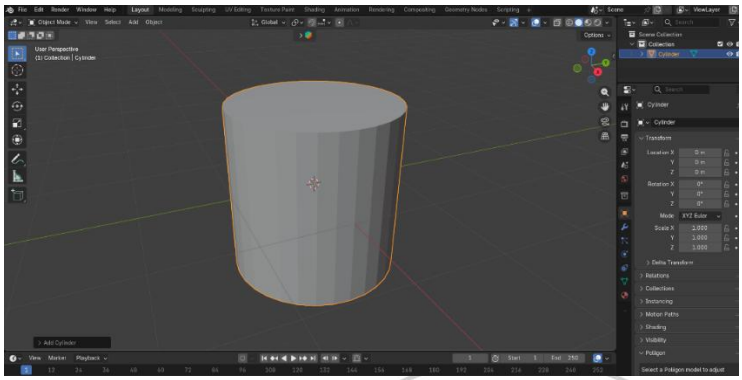
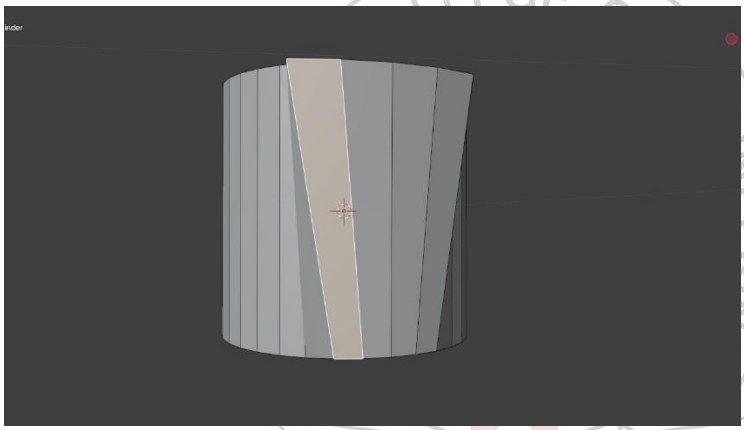
Used tools:

- Extrude (E)
- Scale (S)
- Loop Cut (Ctrl + R)



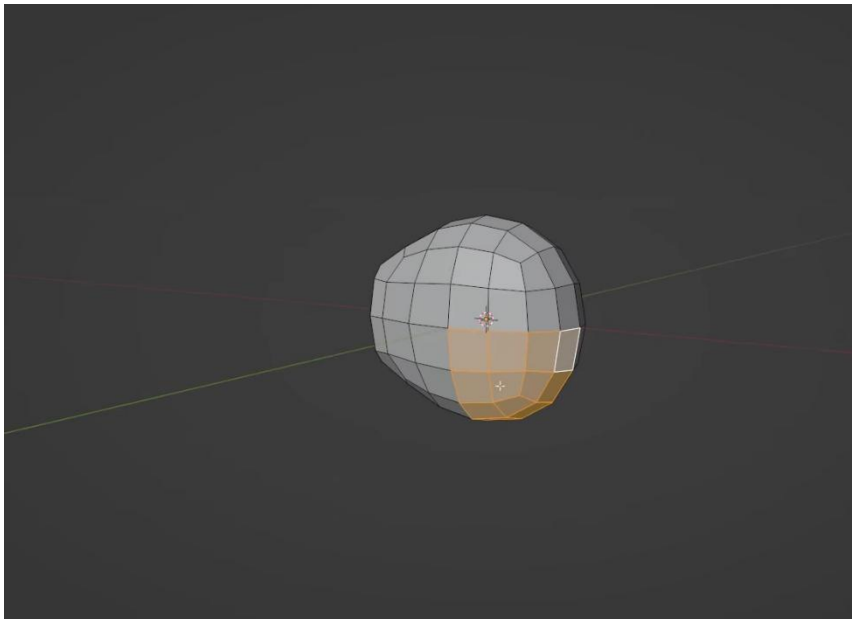


Step 4: Shape the Earbud Head

- Add Sphere
- Delete half and mirror
- Adjust curvature
- Join with stem

Used:

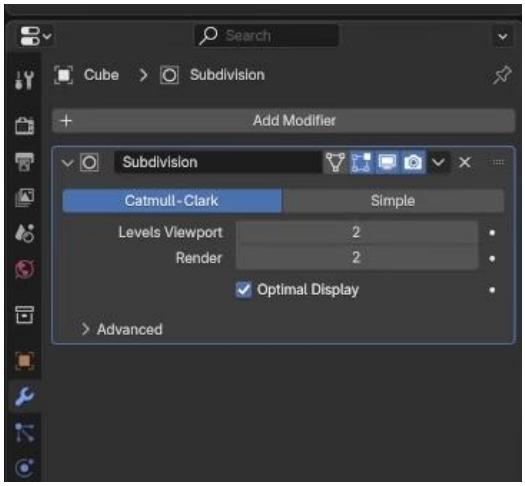
- Proportional Editing
- Smooth shading



Step 5: Apply Modifiers

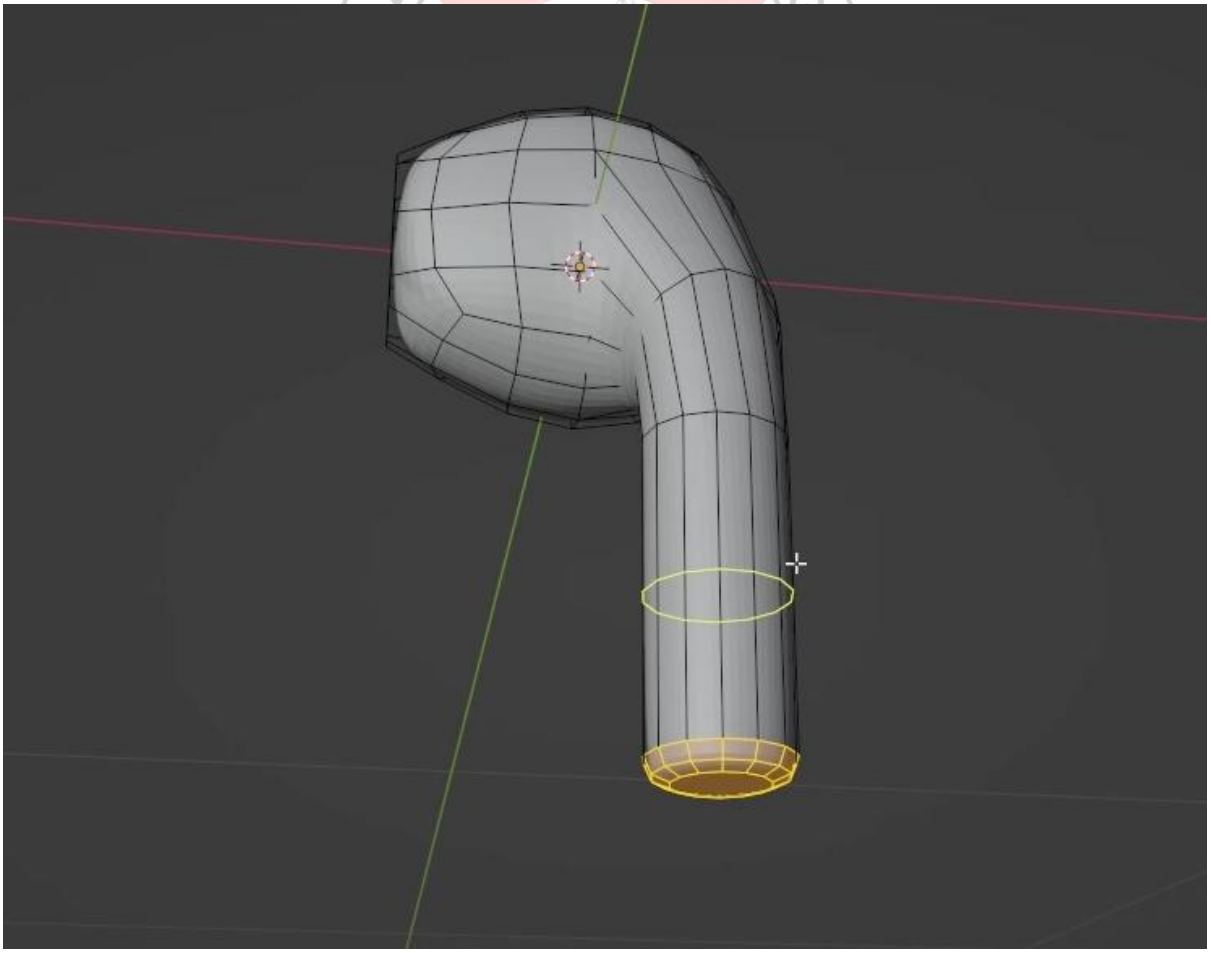
Added:

- Subdivision Surface (Level 2–3)
- Bevel Modifier (small value)
- Mirror Modifier (if symmetrical)



Step 6: Clean Topology

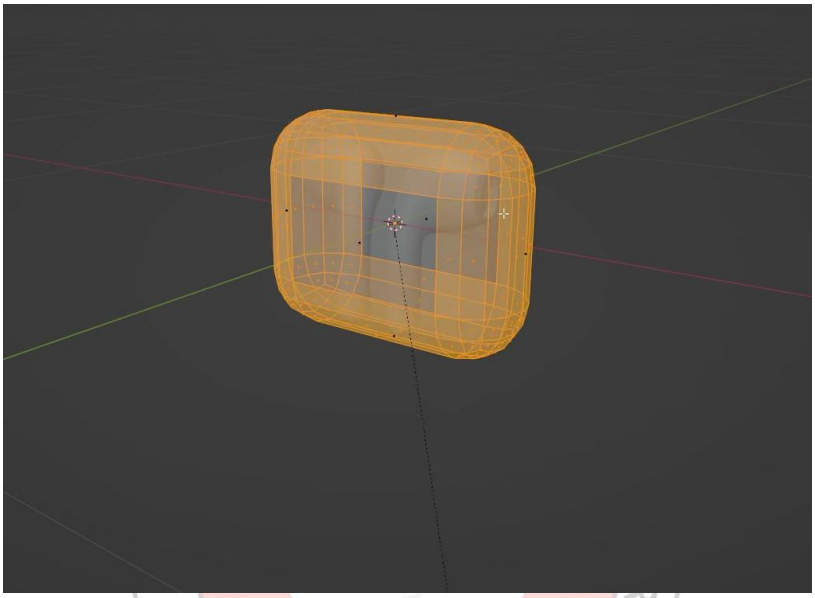
- Adjust edge loops
- Remove unnecessary vertices
- Maintain quad geometry



3. MODELING THE CHARGING CASE

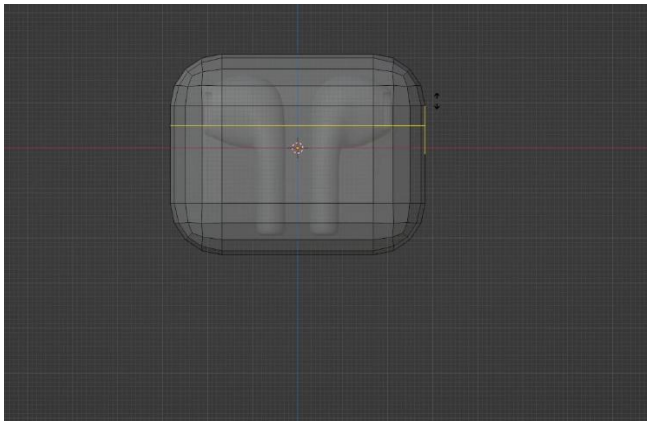
Step 7: Base Case Creation

- Add Cube
- Scale to approximate case size
- Add Loop Cuts
- Bevel edges



Step 8: Hollow Interior

- Duplicate top face
- Extrude inward
- Create inner cavity



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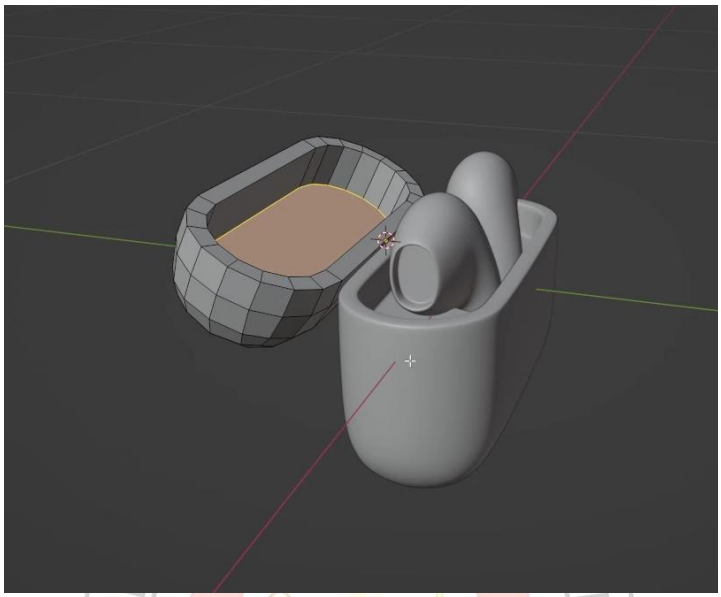
Step 9: Create Lid

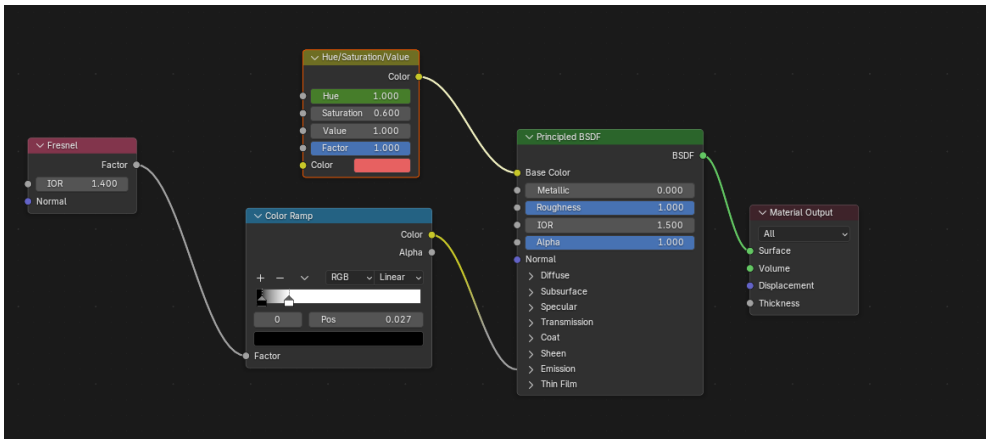
- Separate top faces
- Adjust pivot point
- Add rotation animation pivot



4. SHADING & TEXTURING

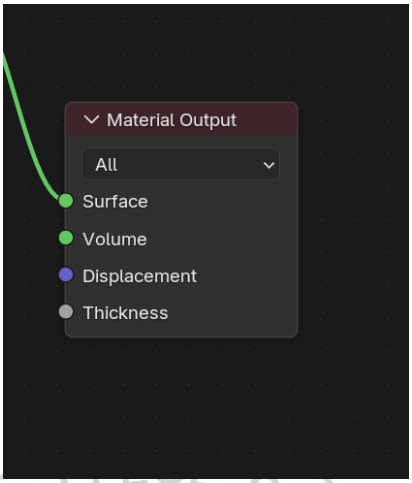
Step 10: Create Plastic Material

- Open Shader Editor
- Add Principled BSDF
- Adjust Roughness (0.2–0.4)
- Slight specular



Step 11: Metallic Ring Material

- Metallic = 1
- Roughness = 0.1–0.2



Step 12: Assign Materials

- Select faces
- Assign specific material slots

5. LIGHTING SETUP

Step 13: Add Key Light

- Add Area Light
- Position 45° angle
- Adjust power

Step 14: Add Fill Light

- Lower intensity
- Opposite side

Step 15: Add Rim Light

- Back side
- Creates edge highlight

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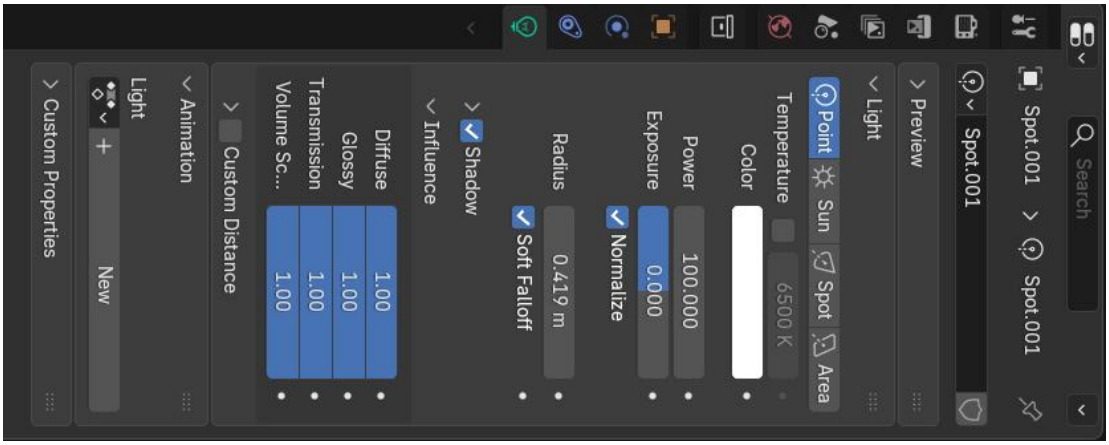
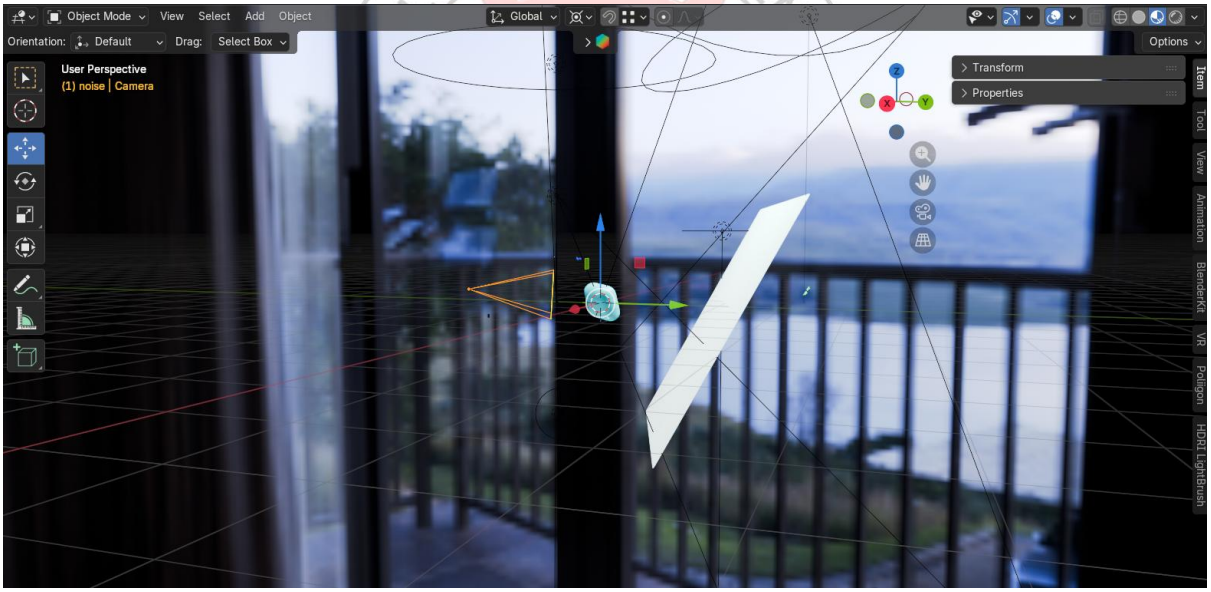
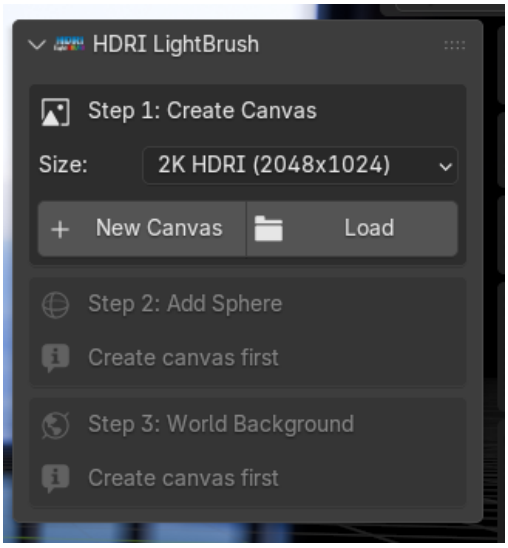


Step 16: Add HDRI

- Go to World Shader
- Add Environment Texture
- Load HDRI

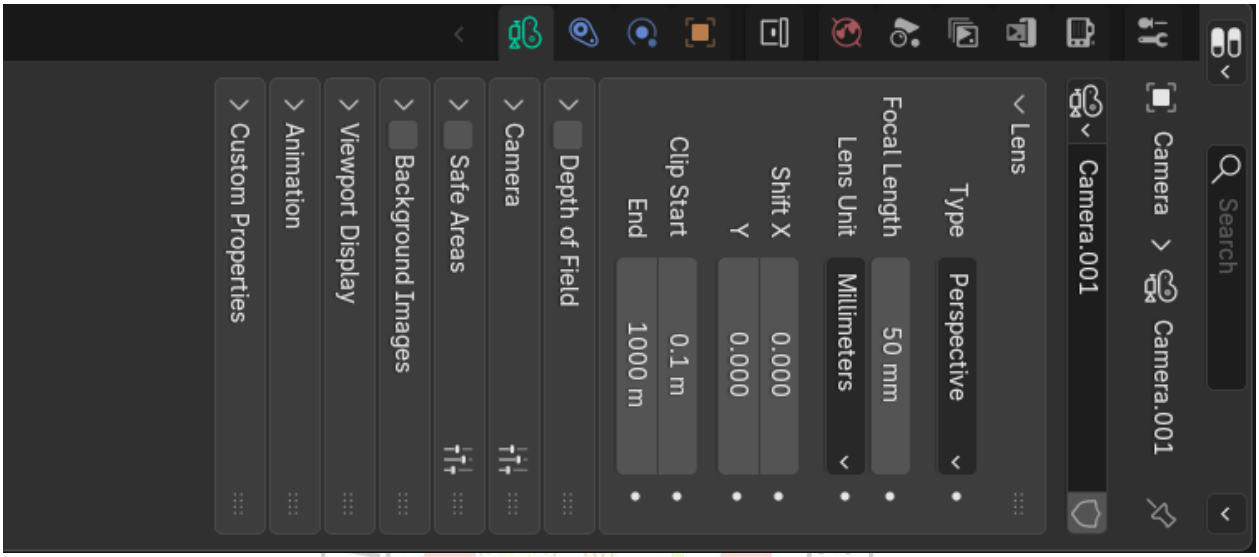




6. CAMERA & ANIMATION

Step 17: Camera Setup

- Add Camera
- Adjust focal length (50–85mm)
- Enable Depth of Field

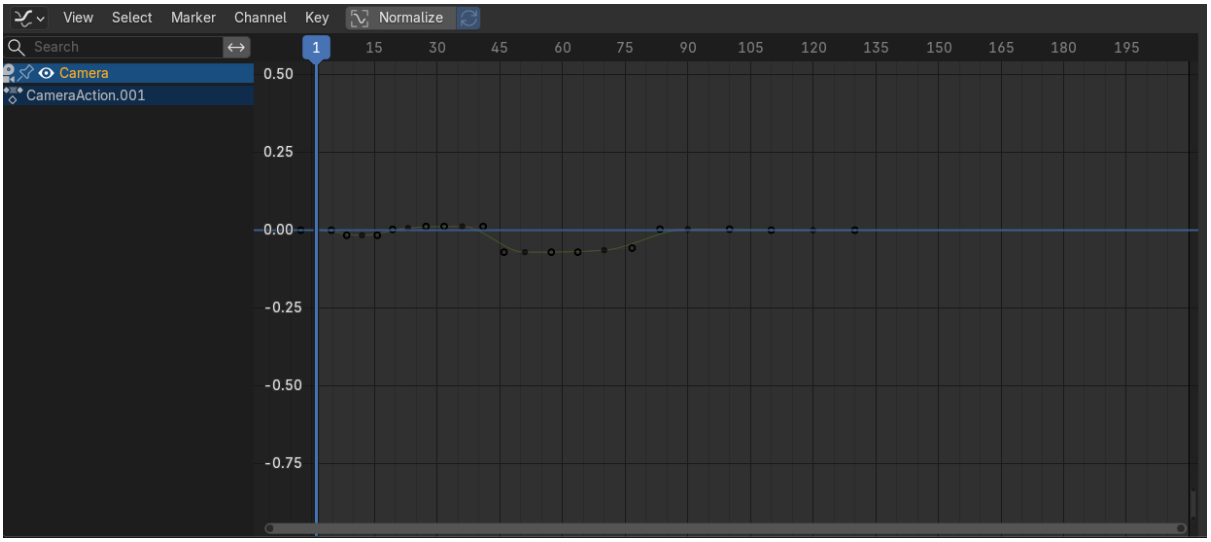


Step 18: Orbit Animation

- Add Empty at center
- Parent camera to empty
- Rotate empty

Step 19: Keyframe Animation

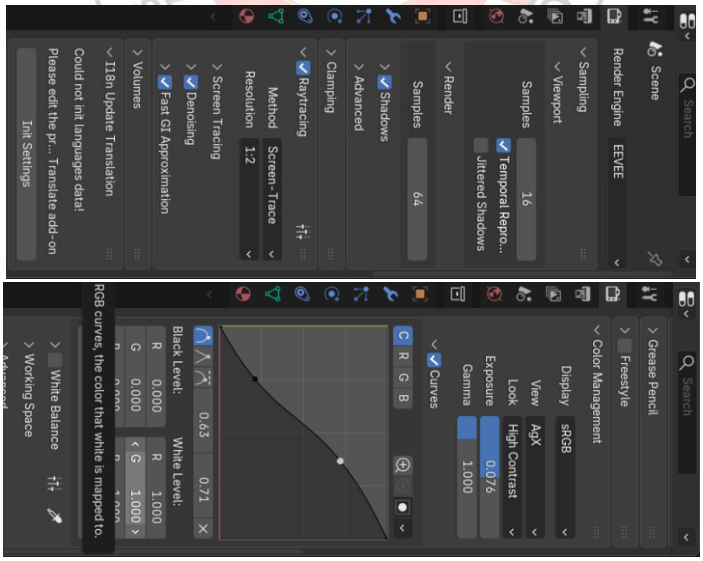
- Insert Location & Rotation keyframes
- Adjust in Timeline



7. RENDERING

Step 20: Render Settings

- Engine: EEVEE
- Samples: 16
- Resolution: 1920x1080
- Denoising Enabled



Step 21: Test Render

- Render Image
- Check noise
- Adjust lighting

8. POST-PROCESSING

Step 22: Compositor Setup

- Enable Use Nodes
- Add Color Balance
- Add Glare Node
- Adjust Exposure

Step 23: Final Output

- Render Animation
- Export as MP4
- 24 fps

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FINAL RESULT

The complete process demonstrated:

Modeling → Shading → Lighting → Animation → Rendering → Compositing → Final Advertisement Output

CHAPTER 11:

PROJECT FILE ACCESS AND REPOSITORY DETAILS

11.1 GitHub Repository Access

The complete project files including:

- Blender (.blend) file
- Texture assets
- HDRI files
- Final rendered video
- Project screenshots
- Project report (PDF)

have been uploaded to a GitHub repository for direct access and verification.

Repository Link:	https://github.com/ketananand76/Minor_Project_7th-sem
Final Output Link:	https://drive.google.com/file/d/1eW85iAvMsx-i-uy2qU0h6Rs0wszxfe4D/view?usp=sharing

SCAN ME

The repository allows reviewers to:

- View project structure
- Access source files
- Verify practical implementation
- Download assets for reference

This ensures transparency and authenticity of the project work.

CHAPTER 12:

REFERENCES

12.1 Online Learning Resources

1. Polygon Runway. (Year). *Wireless Pods Blender Animation Tutorial*. YouTube. Available at: https://youtu.be/Vr_RdSaUSvU?si=ERuRS3cWKulApYy7

2. Blender Official Documentation.
   Available at: https://drive.google.com/file/d/1eW85iAvMsx-i-uy2qU0h6Rs0wszxfe4D/view?usp=sharing

3. Blender Foundation. (2024). *EEVEE Render Engine Documentation*.

4. Randered Mp4 File:- https://drive.google.com/file/d/1eW85iAvMsx-i-uy2qU0h6Rs0wszxfe4D/view?usp=sharing

12.2 Software References

- Blender Version 5.0
- HDRI Environment from Poly Haven (if used)

HDRI File:-

wooden_studio_08_8k
.hdr

https://polyhaven.com/a/wooden_studio_08

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